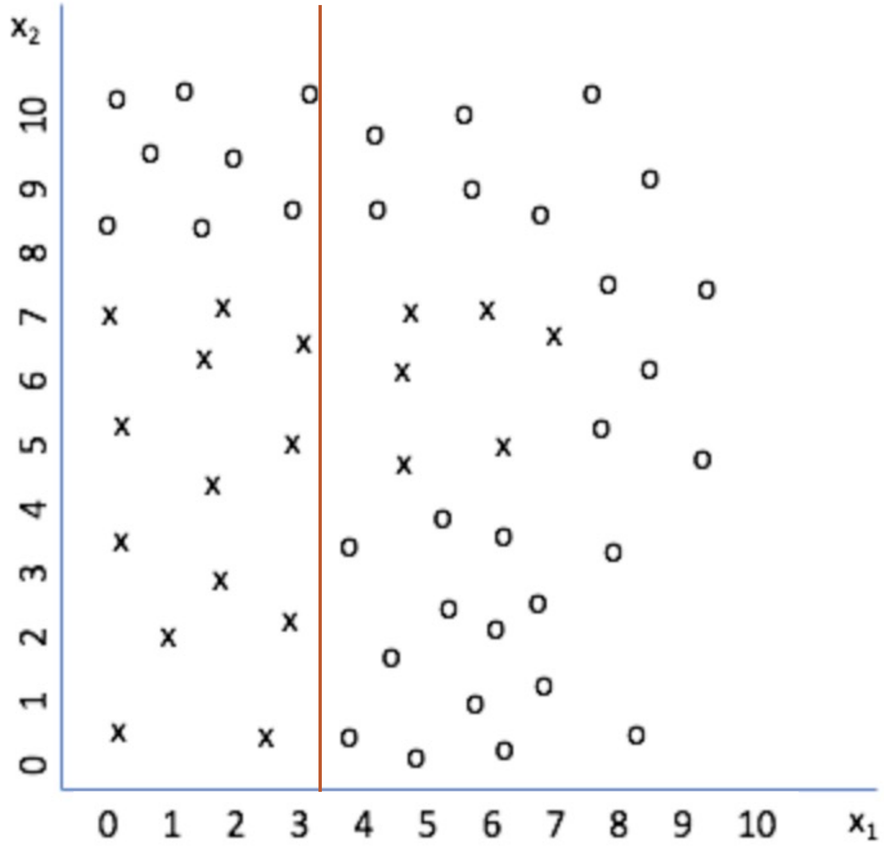
Lecture 6: Generalization and Overfitting & Exam Review

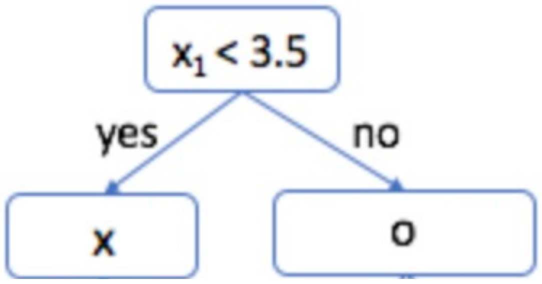
Generalization and Overfitting

- Generalization is the property of a model or modeling process, whereby the model applies to data that were not used to build the model.
- Overfitting is the tendency of data mining procedures to tailor models to the training data, at the expense of generalization to previously unseen data points.

Generalization and Overfitting

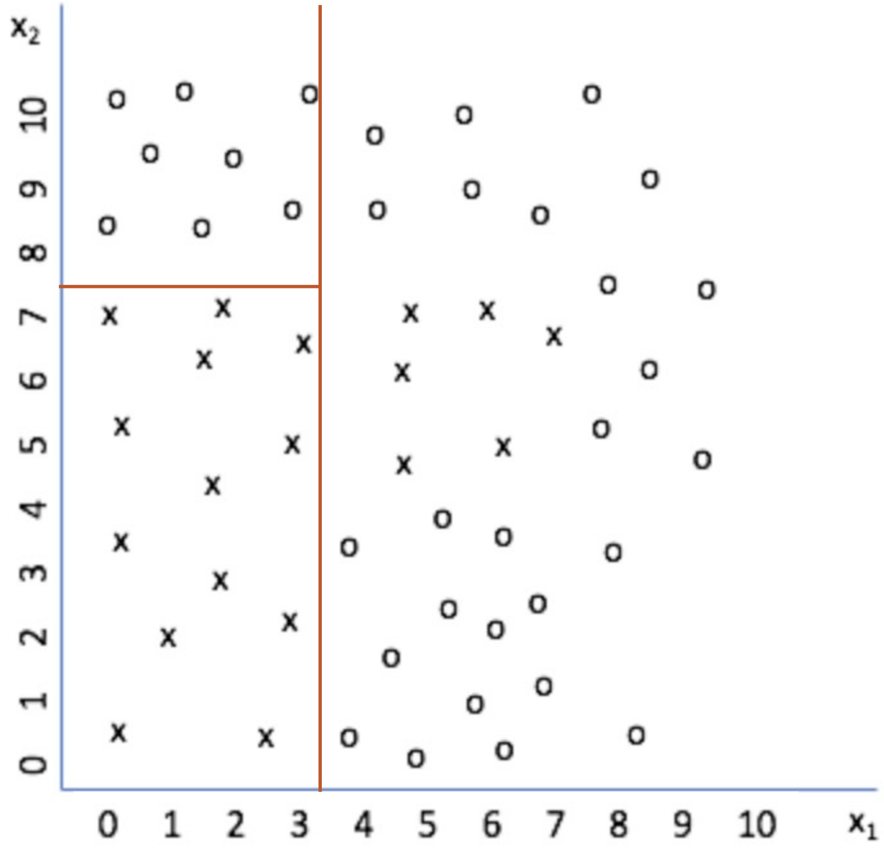
- Decision Tree

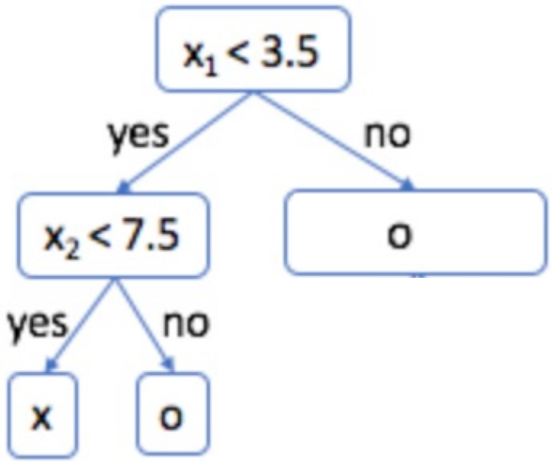




Generalization and Overfitting

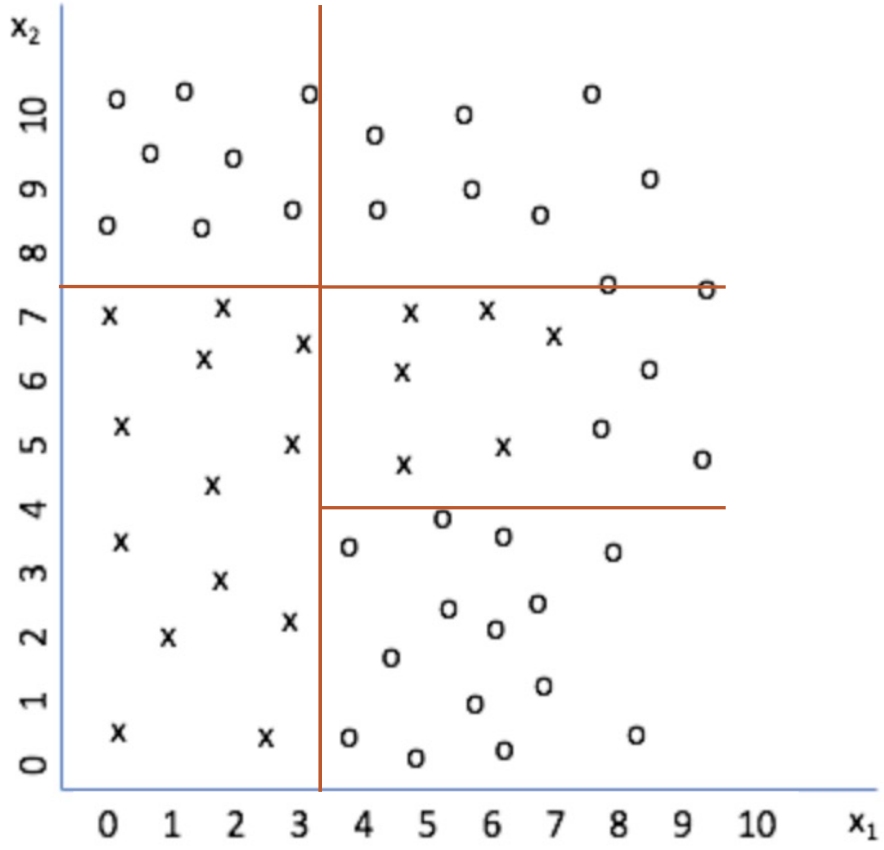
- Decision Tree

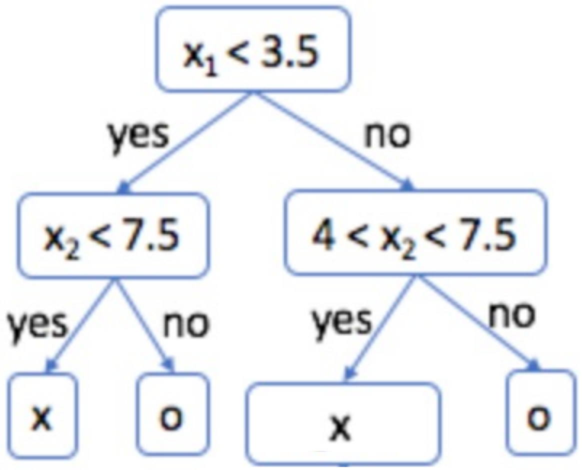




Generalization and Overfitting

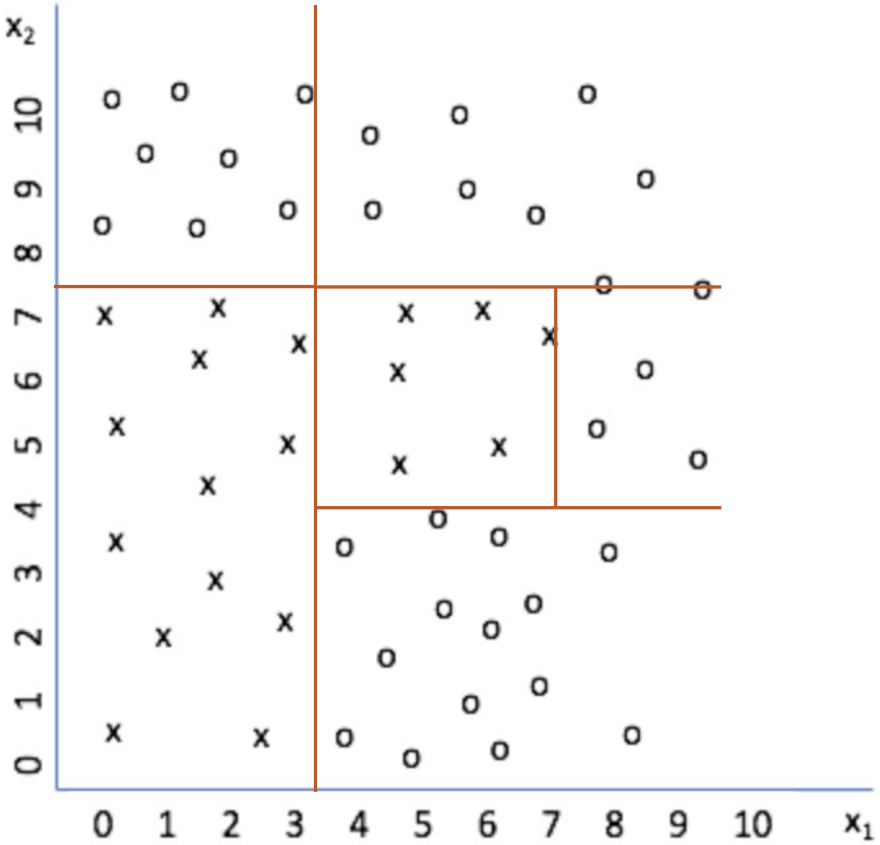
- Decision Tree

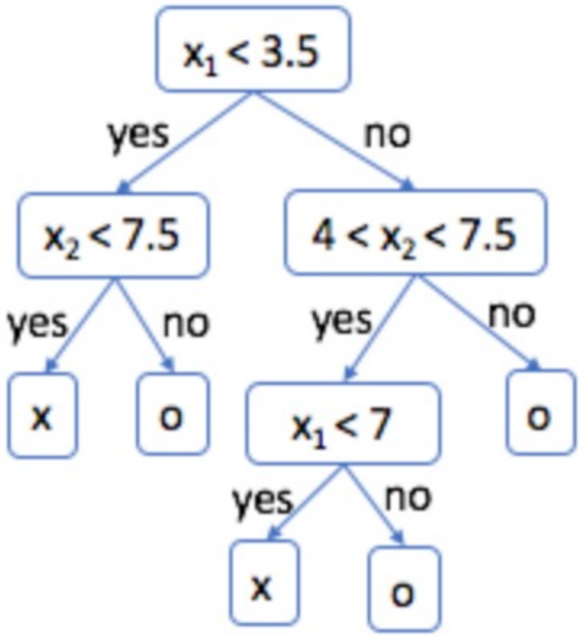




Generalization and Overfitting

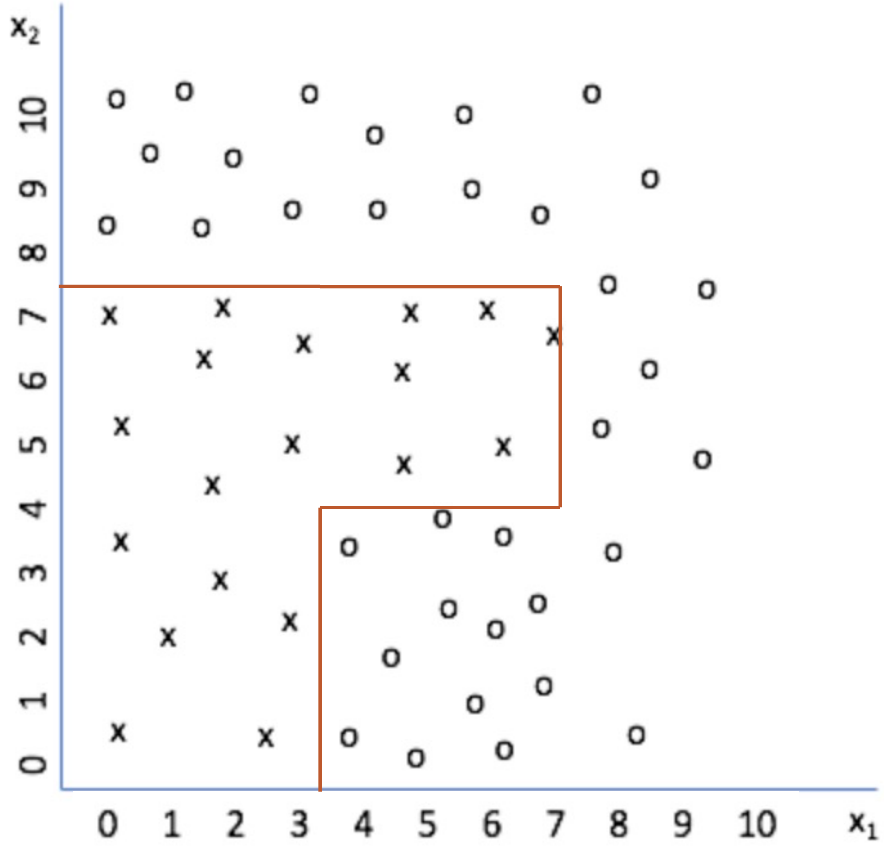
- Decision Tree

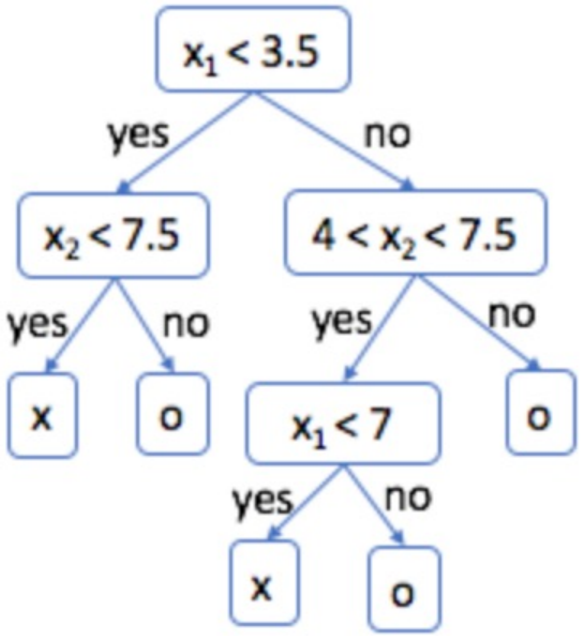




Generalization and Overfitting

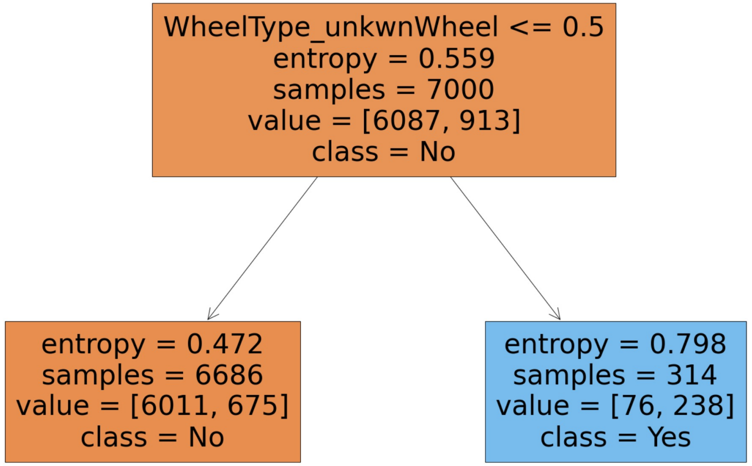
- Decision Tree





Generalization and Overfitting

- Comparing the model **performances** on **training** and **testing** data
  - Max depth = 1



Training performance

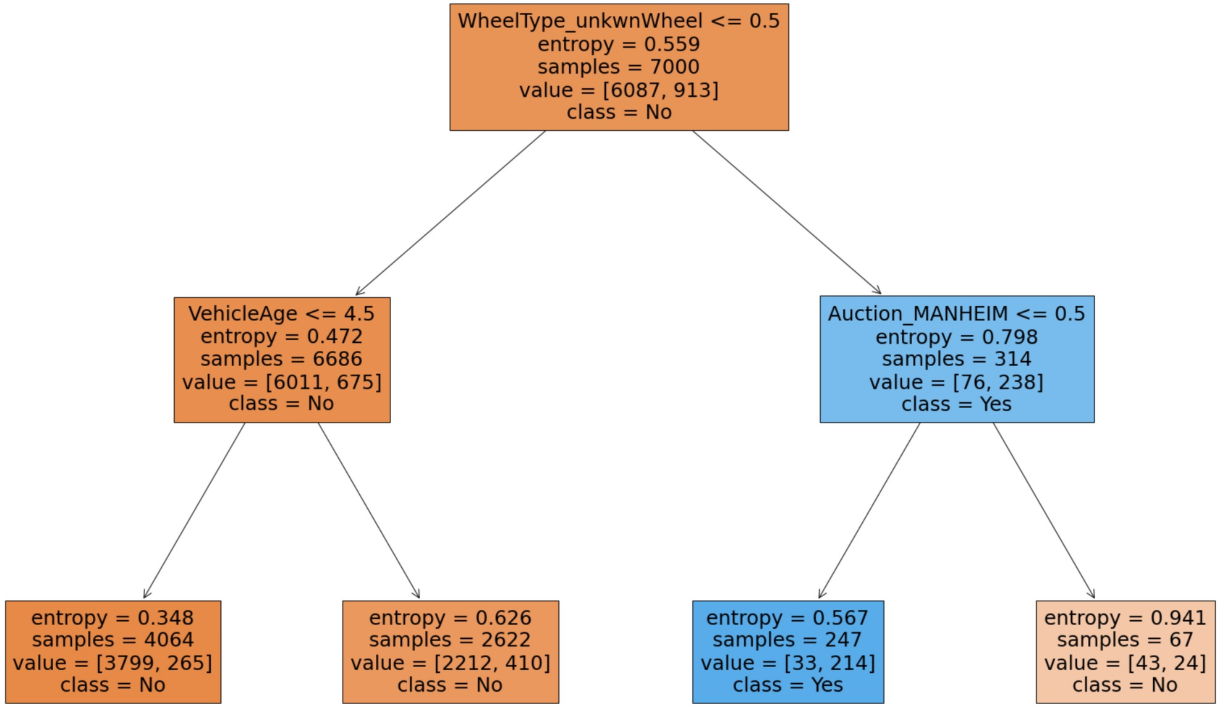
	precision	recall	f1-score	support
No	0.90	0.99	0.94	6087
Yes	0.76	0.26	0.39	913
accuracy			0.89	7000
macro avg	0.83	0.62	0.66	7000
weighted avg	0.88	0.89	0.87	7000

Testing performance

	precision	recall	f1-score	support
No	0.90	0.99	0.94	2618
Yes	0.76	0.26	0.39	382
accuracy			0.90	3000
macro avg	0.83	0.62	0.67	3000
weighted avg	0.88	0.90	0.87	3000

Generalization and Overfitting

- Comparing the model **performances** on **training** and **testing** data
  - Max depth = 2



Training performance

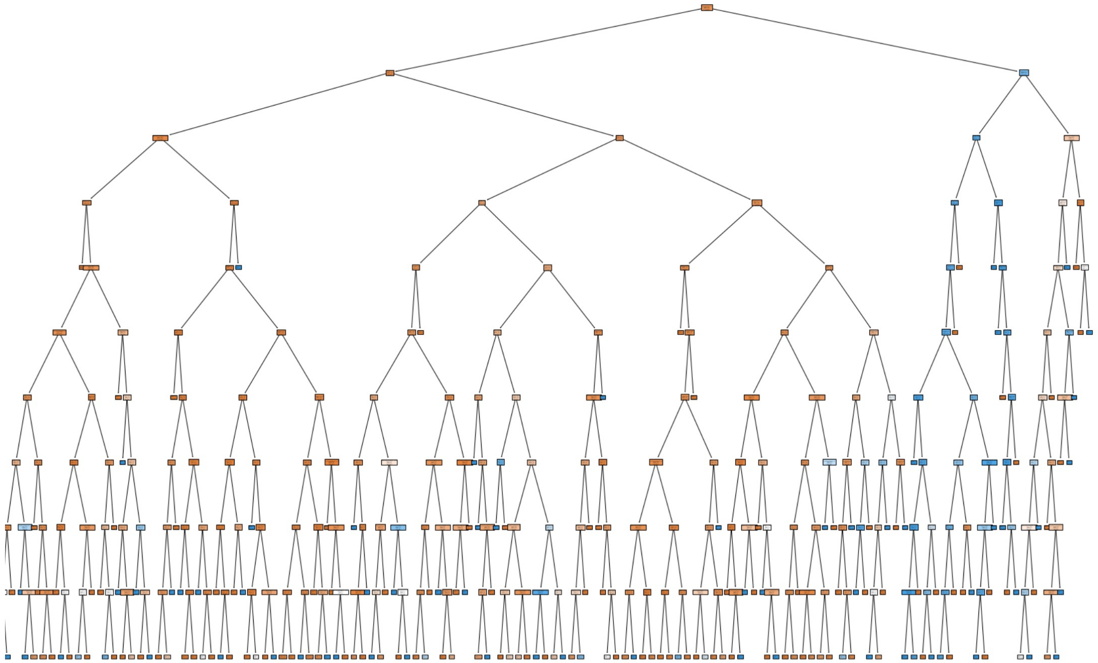
	precision	recall	f1-score	support
No	0.90	0.99	0.94	6087
Yes	0.87	0.23	0.37	913
accuracy			0.90	7000
macro avg	0.88	0.61	0.66	7000
weighted avg	0.89	0.90	0.87	7000

Testing performance

	precision	recall	f1-score	support
No	0.90	1.00	0.94	2618
Yes	0.90	0.22	0.36	382
accuracy			0.90	3000
macro avg	0.90	0.61	0.65	3000
weighted avg	0.90	0.90	0.87	3000

Generalization and Overfitting

- Comparing the model **performances** on **training** and **testing** data
  - Max depth = 10



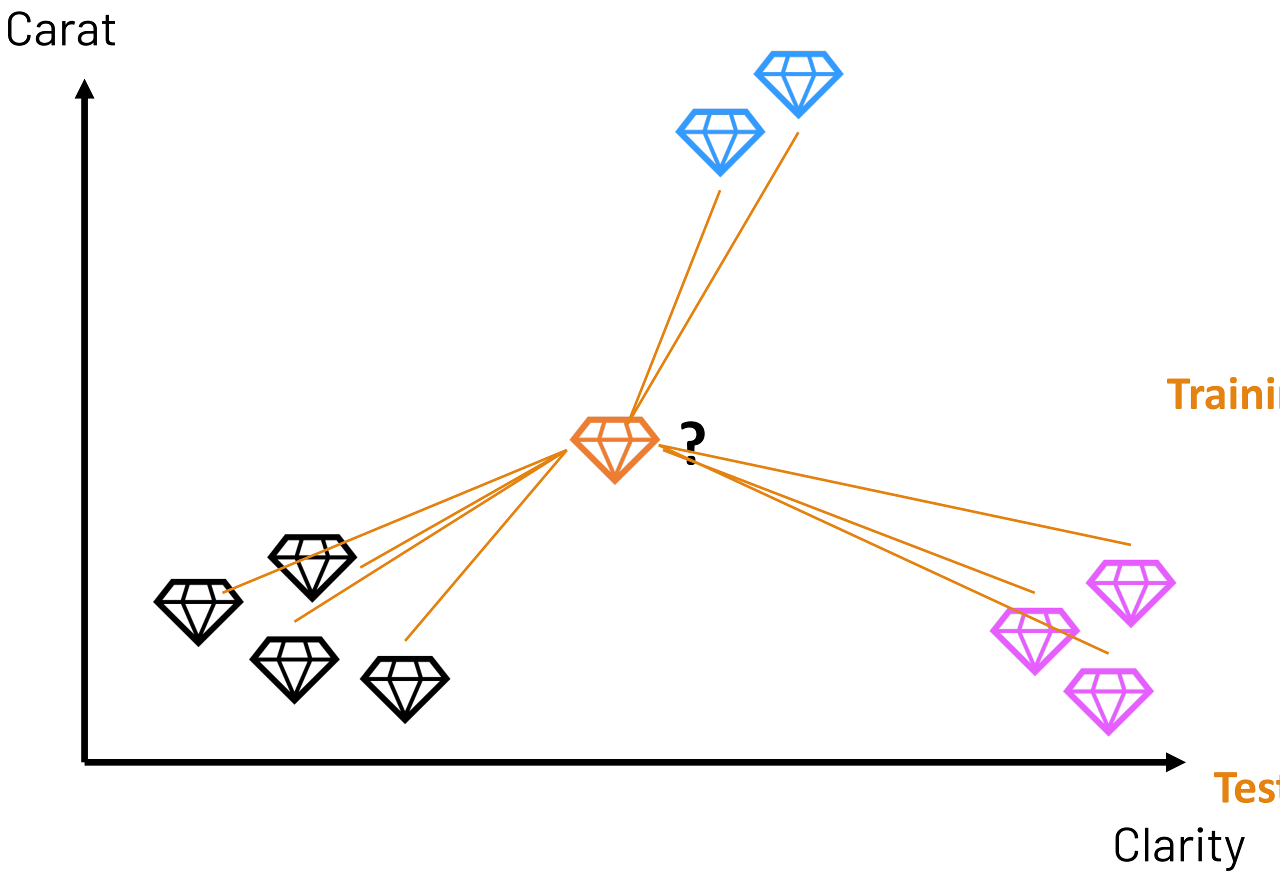
Training performance

	precision	recall	f1-score	support
No	0.92	1.00	0.96	6087
Yes	0.94	0.41	0.57	913
accuracy			0.92	7000
macro avg	0.93	0.70	0.76	7000
weighted avg	0.92	0.92	0.90	7000

Testing performance

	precision	recall	f1-score	support
No	0.90	0.97	0.93	2618
Yes	0.56	0.25	0.35	382
accuracy			0.88	3000
macro avg	0.73	0.61	0.64	3000
weighted avg	0.86	0.88	0.86	3000

Nearest Neighbor



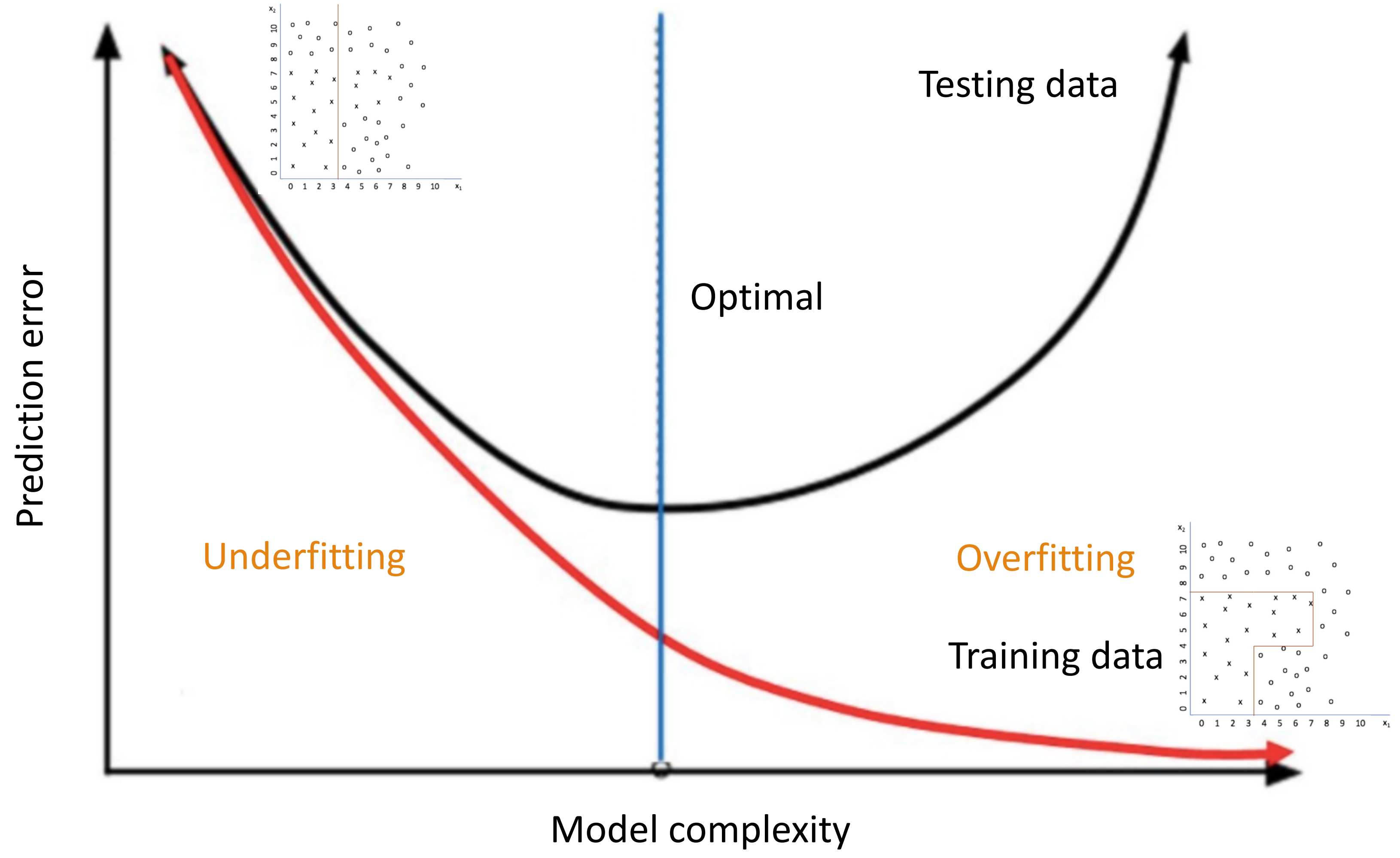
Training data

Testing data

	Clarity	Carat	Class	Distance
Diamond1	0.778	1	A	0.747
Diamond2	0.667	0.92	A	0.618
Diamond3	1	0.2	B	0.696
Diamond4	0.944	0.12	B	0.672
Diamond5	0.889	0	B	0.685
Diamond6	0.222	0	C	0.415
Diamond7	0.111	0.2	C	0.299
Diamond8	0.111	0	C	0.458
Diamond9	0	0.08	C	0.462
Diamond10	0.333	0.4	???	0

What is the largest/smallest *k* value?

Generalization and Overfitting



Generalization and Overfitting

Causes for Overfitting:

- Training data is not a good representation of testing (new) data
  - Insufficient training data
  - Noises in data: inconsistent class labels for the same values in feature set (input attributes)
  - Outliers in data: the number of samples with a given combination of class labels and feature values is small.
- An algorithm's inability to avoid overfitting noises/outliers or to train generalizable models via small amounts of training data
  - Complex model

Generalization and Overfitting

Avoidance of Overfitting

- Data strategies
  - Secure sufficient data
  - Identify and handle potential outliers and noises
- Evaluation strategies
  - **Identifying overfitting**: comparing the model **performance** on **training** and **testing** data
    - **Avoid big gap between the training and testing accuracy**
- Model strategies
  - Select proper algorithm and manage model complexity
    - Compare different algorithms
    - Lower model complexity via method-specific parameters